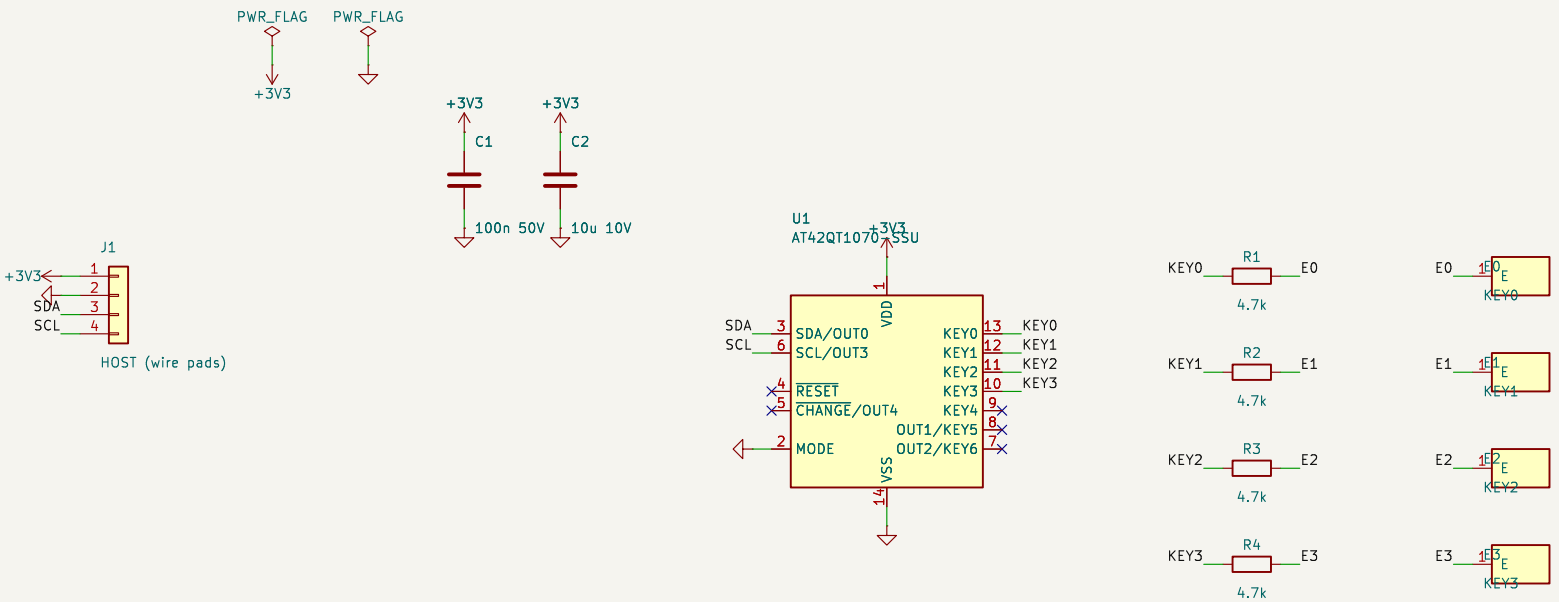
TOUCH BOARD rev D (decision 5A): 4 electrodes + AT42QT1070, one HOST wire-pad set; replaces rev C in the chin seat



AT42QT1070 (Atmel 9596C-AT42-05/2013): key outline 6 x 6 mm or larger (panel thickness dependent); plastic panels up to 5 mm;
Cx 1-30 pF per key; MODE = GND selects comms mode (rev A root cause); no pull-ups here (host 1.8 k on the Pi).
Electrodes 16 x 16 mm (rev C: Ø12); QTAN0062 2.3 - electrode dimension >= 4 x panel thickness; panel = 2.5 wall + 0.7 cloth
= 3.2 mm (rev F model, not measured) -> >= 12.8 mm. Centres unchanged from rev F: front-view X 9/33/57/81, Y 13.
Board 90 x 44 x 1.6 (rev F seat), no holes except tented vias; front = electrodes only; all parts and the HOST pads on the back.
BME680 / VEML6030 leads now land on the main board (J10); VL53L0X cluster dropped.

Replaces hardware/rev-c (unchanged, kept for reference)
Decision 5A (Daniel 2026-11-07): same 90 x 44 outline, electrodes + AT42QT1070 + HOST pads
KiCad 9.0.9 - generated by hardware/main-board/gen/touch_board.py
HA panel (neocleous/ha-panel)

Sheet: /
File: touch-board.kicad_sch

Title: HA panel touch board rev D

Size: A4	Date: 2026-11-07	**Rev: D**
KiCad E.D.A. 9.0.9		Id: 1/1